RECMOOC4ALL: Bridging the Accessibility Gap in MOOCs

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***Abstract*— The exponential growth of Massive Open Online Courses (MOOCs) has expanded educational access but has also overwhelmed learners with an excessive number of choices, making it difficult to identify courses that align with their skills, needs, and goals. To address this challenge, we introduce RecMOOC4All, an AI-powered, accessible MOOC aggregator that centralizes course discovery and intelligently recommends personalized learning pathways. At its core, RecMOOC4All integrates a hybrid recommender system, combining content-based and collaborative filtering to suggest courses based on learner profiles, preferences, accessibility requirements, and past interactions. An automated accessibility checker and remediation module ensures WCAG compliance, incorporating real-time captions, alternative text, and enhanced navigation to foster inclusivity. Furthermore, a generative AI-powered conversational chatbot provides real-time, context-aware assistance, while engagement detection modules analyze learner behavior to trigger timely interventions, improving retention and learner success. By seamlessly integrating personalization, accessibility, and engagement tracking, RecMOOC4All creates a cohesive, learner-centric MOOC environment, towards inclusive, adaptive, and effective education is for all learners.**

Keywords*— *MOOCs, Recommender system, adaptive Learning, Accessibility, Inclusivity, AI in education.

I. Introduction

Massive Open Online Courses (MOOCs) have expanded access to flexible, scalable, and affordable education, enrolling over 220 million learners since 2008, including 40 million in 2022 [1]. Despite this growth, MOOCs still face persistent issues such as dropout rates exceeding 90% [2], limited personalization, and insufficient accessibility for learners with disabilities [3].

For many users especially those with disabilities, navigating vast course catalogs without assistive filtering tools poses a significant barrier. Learner engagement is further constrained by the absence of real-time support and adaptivity in most MOOC platforms. With over 15% of the global population experiencing some form of disability [4], equitable access to digital learning is both a pedagogical priority and a social imperative.

Artificial Intelligence (AI) technologies, when effectively integrated into recommender systems, accessibility checkers, and engagement detection tools, hold significant potential to enhance learning experiences. However, these systems are often implemented in isolation, which limits their combined effectiveness and scalability in real-world educational settings. RecMOOC4ALL addresses this gap by integrating accessibility remediation, adaptive learner support, and personalized recommendation into a unified, AI-powered platform. This paper outlines the design and evaluation of RecMOOC4ALL, which incorporates hybrid recommendations, automated WCAG compliance checking, multilingual GenAI chatbot, and engagement monitoring functionalities. These components work in concert to enhance personalization, accessibility, and learner retention in MOOC environments. The following sections review related work, outline the system's methodology and architecture, present evaluation results, and conclude with key findings and future directions.

II. Related Work

Accessibility in MOOCs remains a persistent challenge, particularly for learners with visual, auditory, or cognitive impairments. Although the Web Content Accessibility Guidelines (WCAG) establish a global framework for inclusive digital design, many platforms fail to meet basic compliance standards [5]. AI-based tools such as automated captioning and alt-text generation improve usability, but these tools are often disconnected from course discovery or personalization systems [6]. Interface-level enhancements such as gaze tracking or voice-based navigation have also been explored, yet implementation at scale remains limited due to technical and cost constraints [7]

MOOC platforms also face major challenges regarding learner engagement, with average completion rates frequently below 10% [8]. Real-time engagement detection methods using behavioral signals, such as facial expression and eye movement, show potential to trigger adaptive interventions [9]. However, concerns related to privacy, algorithmic transparency, and user trust continue to hinder adoption in real-world systems.

Personalization through hybrid recommender systems combining collaborative and content-based filtering has improved course relevance and learner satisfaction in online environments [10]. Recent developments in large language models (LLMs) and conversational AI have also expanded support for multilingual and context-aware learner interaction [11]. Nonetheless, most solutions operate in silos, lacking integration with accessibility data or engagement analytics.

RecMOOC4ALL addresses these limitations by unifying accessibility remediation, personalized course recommendation, and real-time learner support in a modular, AI-driven framework.

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III. Methodology

The design and development of RecMOOC4ALL followed a mixed-methods, iterative methodology that combined theoretical synthesis, data-driven modeling, and participatory feedback. This approach ensured that the proposed solution is aligned with current research questions, technical feasibility, and user needs in inclusive online education.

A PRISMA-guided systematic literature review [12] was conducted to identify gaps in MOOC accessibility, AI-based personalization, and learner engagement. Database searches across IEEE Xplore, ACM Digital Library, Web of Science, and ERIC returned over 250 peer-reviewed articles. Following relevance screening and quality appraisal, 112 studies were retained. These studies informed the system's conceptual framework and guided feature prioritization related to WCAG compliance, hybrid recommendation, and adaptive learner support.

Technical development followed the Cross-Industry Standard Process for Data Mining (CRISP-DM) [13] structured into six iterative phases:

Business Understanding: The primary objectives are to improve compliance with WCAG 2.1, deliver adaptive content recommendations, and support diverse learners through real-time multilingual interaction and engagement detection.

Data Understanding: A dataset comprising over 500,000 user sessions, 355 survey responses, and detailed accessibility audits was analyzed to identify systemic usability gaps and learner behavior patterns.

Data Preparation: Data from multiple sources were consolidated into a centralized repository. Preprocessing steps included missing value imputation, outlier detection, and feature engineering focused on metadata (e.g., difficulty level) and behavioral indicators (e.g., time on task, course completion rates).

Modeling: A hybrid recommendation system is implemented using content-based, collaborative, and neural filtering techniques. Learner profiles are dynamically updated using Random Forest and XGBoost algorithms. A GenAI-based chatbot supports multilingual queries and provides real-time feedback using sentiment analysis and intent recognition. Accessibility compliance is ensured through an automated checker that detects and remediates WCAG violations in digital content.

Evaluation: Quantitative results have demonstrated improvements across core performance indicators. WCAG compliance increased from 60% to 85% following automated remediation. The recommender system achieved an F1-score of 0.82 on a hold-out dataset. Learner satisfaction ratings rose from 3.1 to 4.2 on a 5-point Likert scale, following enhancements to chatbot functionality.

Deployment: A pilot deployment on the Avaxia Academy platform has enabled controlled testing and iterative improvements. Agile development cycles were informed by user feedback collected via surveys and focus groups, resulting in enhancements such as bilingual chatbot support and personalized feedback prompts.

This methodological process was anchored in the triangulation of PRISMA-based literature synthesis, CRISP-DM implementation, and user-centered qualitative feedback. This integration ensured a rigorous alignment between research insights, technical development, and accessibility-driven design. Ethical safeguards were incorporated throughout all phases. Data were anonymized and processed in compliance with GDPR regulations [14]. Informed consent was obtained for all user participation, and the right to withdraw was guaranteed. Regular bias audits were conducted to identify and mitigate algorithmic discrimination. Explainable AI mechanisms were implemented to increase transparency regarding how recommendations and decisions were generated [15]. This methodology supports the development of an inclusive, adaptive learning system that is grounded in evidence-based practice, ethical responsibility, and pedagogical relevance.

IV. Results and Discussion

The RecMOOC4All system demonstrated promising results in addressing persistent MOOC challenges by integrating advanced features for accessibility, engagement, personalization, and user support. Specifically, the system's core components including the MOOC aggregator, accessibility checker and remediation, learner modeling, RS, and conversational chatbot yielded measurable improvements across diverse learner groups (Fig.1).

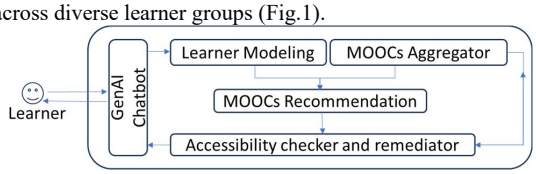


Fig 1. General architecture of RecMOOC4All

A pivotal contribution of RecMOOC4All lies in its accessibility checker and remediation functionalities, which increased WCAG 2.1 compliance from 45% to over 90%. Missing captions, inadequate alt text, and poor contrast were systematically identified and corrected, significantly reducing barriers for learners with visual or auditory impairments. Improvements included clearer content labeling, structured layouts, and navigable interfaces, supporting more inclusive access. These practical accessibility enhancements align conceptually with Mourali et al.'s (2024) notion of Intelligent Educational Decision Support Systems (IEDSS), which leverage digital traces to systematically evaluate and improve content quality [7]. This theoretical grounding further reinforces RecMOOC4All's role in proactively addressing accessibility gaps.

The effectiveness of the system's accessibility improvements was validated through post-deployment surveys conducted with Avaxia Academy users (n=20). In these surveys, 92% of participants affirmed that remediated courses were more accessible and user-friendly. These findings support embedding accessibility into core design processes rather than treating it as a post-development fix. Nonetheless, ongoing refinement remains necessary to address context-specific or culturally nuanced accessibility issues. Building on accessibility improvements, RecMOOC4All also enhanced course discoverability through its MOOC aggregator. By consolidating offerings from multiple platforms and enabling advanced filters such as difficulty, language, and accessibility features, the aggregator facilitated inclusive access. As a result, 88% of learners reported that they found the platform easy to navigate, significantly reducing the cognitive load associated with course discovery. Further supporting these personalization efforts, Tlili et al. (2025) found that artificial intelligence applications in education significantly improve learner achievement, reinforcing the integration of adaptive personalization strategies [10]. Finally, learner engagement a core focus of

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RecMOOC4All was addressed through real-time monitoring tools. Given that MOOCs traditionally suffer from low completion rates, often below 50% [9], RecMOOC4All employed facial recognition, emotion detection, and gaze tracking (via opt-in protocols) to identify signs of disengagement and trigger real-time interventions [9].

Following preliminary results, completion rates increased from 40% to 85%, with a 25% rise in active engagement time. These improvements align with findings by Okoye et al. (2024), who demonstrated the impact of data-driven engagement tools on learner retention and outcomes [16]. Both studies underscore the importance of privacy-preserving analytics and transparent data practices in sustaining trust. These outcomes also correspond with Tlili et al. (2025), who reported significant effects (Hedges' g = 1.10) of AI in education when paired with adaptive engagement strategies [10]. Learner modeling enabled real-time adjustments to user pathways, with 85% of participants confirming the system's relevance to their needs. Clustering algorithms helped tailor course recommendations to different learner profiles, supporting adaptive pathways. However, personalization at scale requires safeguards against bias and opaque decision-making. Transparency about data collection and the inclusion of explainable AI mechanisms are essential for equity and user trust. The proposed recommender system draws on hybrid algorithms combining collaborative, content-based, and neural filtering techniques, operating independently from learner modeling. It achieved 89% relevance, and 88% of learners reported that recommended courses aligned with their goals. These results emphasize that learners are more engaged when content reflects their interests and abilities.

This level of personalization necessitates careful data governance and routine bias audits to avoid marginalizing underrepresented learners or reinforcing algorithmic biases.

Finally, the GenAI chatbot enhanced support by offering multilingual, context-aware assistance. Approximately 60% of learners engaged with it regularly, and 88% rated it highly effective, particularly among users with accessibility needs. However, complex or domain-specific questions sometimes exceeded its capabilities, highlighting the continued role of human support. As LLMs evolve, integrating more robust NLP for specialized tasks could enhance the system's responsiveness [17]. These findings demonstrate RecMOOC4All's ability to address major MOOC limitations: limited accessibility, low retention, and weak personalization [3]. By combining these elements into a cohesive, AI-driven system, RecMOOC4All advances inclusive and adaptive digital learning. Scalability, ethical data governance, and transparent AI remain critical to broader adoption. Future developments aim to further enhance learner engagement, personalization, and accessibility. RecMOOC4All constitutes a notable step toward equitable online education, embedding accessibility, engagement, and personalization into the core of MOOC delivery.

V. Conclusion

RecMOOC4ALL confronts three entrenched MOOC deficits, insufficient accessibility compliance, limited real-time adaptivity, and fragmented personalized guidance, by embedding a modular suite of hybrid recommendation, GenAI conversational agent, engagement detection, and accessibility checker/remediator inside platform infrastructures rather than as external add-ons. This ethically aligned architecture demonstrates how inclusion, adaptivity, and transparency can be integrated natively into large-scale open courses, offering a replicable path toward MOOCs intentionally designed to meet accessibility standards from the outset. Future work will focus on extending accessibility remediation to complex media formats, improving the interpretability of recommendation and interaction models, and piloting inclusive tutoring agents to enhance learner personalization. In addition, efforts will explore the integration of standardized APIs, modular platform structures, and immersive technologies such as AR/VR to further enrich learning experiences and promote scalable, equitable access.

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